ChatGPT

Sourcing and supply chain for Canada’s River-class destroyer program

Background

Canada’s **River-class destroyer (RCD)**—also known as the **Canadian Surface Combatant (CSC)**—will replace the Royal Canadian Navy’s aging destroyers and frigates. Irving Shipbuilding Inc. (ISI) in Halifax is the prime contractor. A March 2025 federal news release explained that the implementation contract is initially worth **$8 billion** and covers construction and delivery of the first three ships as well as training, spares and maintenance packages [1]. The government estimates that building the first three vessels will cost **$22.2 billion** (excluding taxes) and that the program will create or maintain **~5,250 jobs annually** and inject **$719 million** into Canada’s GDP during 2025–2039 [2]. The RCD therefore includes an extensive industrial base; the sections below map the major systems and their sourcing.

Shipbuilding and structural materials

Component	Supplier & manufacturing location	Evidence
Hull steel, aluminum and plate	**Russel Metals** (Canadian metals distributor) will supply steel plates, profiles and aluminum sheets for parts of the ships. Materials are delivered to Irving Shipbuilding and fabricators across Canada [3]. The long-term partnership is expected to benefit Nova Scotia jobs [4].	Russel Metals article on the National Shipbuilding Strategy [3].
Lead ballast	**Marswell Metal Industries** of Burlington, Ontario will manufacture lead ballast components using *100 % Canadian-sourced materials and labour* [5]. These components provide ship stability.	Marswell news release (March 12 2026) [5].
Propellers and fin stabilizers	**Kongsberg Maritime** secured contracts to supply *twin fixed-built propellers*, *non-retractable fin stabilisers*, *steering gear*, *rudders* and *replenishment-at-sea (RAS) high-point equipment*. Deliveries of the propellers for the first three ships start in **2028** [6]. Kongsberg told Janes that propeller hubs and blades are manufactured in Europe, but assembly and final integration occur in Canada [7].	Naval Today (2025) [6] and Janes reporting [7].
Mission Bay Handling System	**Rolls-Royce** supplies an automated Mission Bay Handling System (MBHS). The system, used for launching boats and moving containers, is manufactured at Rolls-Royce’s **Naval Handling Centre of Excellence in Peterborough, Ontario** [8].	Rolls-Royce release (2024) [8].

Component	Supplier & manufacturing location	Evidence
Helicopter handling equipment	**Curtiss-Wright** will integrate its *ASIST* (Aircraft Ship Integrated Secure and Traverse) helicopter handling system under a **US$25 million** contract. The ASIST system automates helicopter landing and securing and will be integrated by Curtiss-Wright's Electro-Mechanical Systems division [9] .	Curtiss-Wright press release [9] .
Engine intake/ exhaust & IR suppression	Ottawa-based **W.R. Davis Engineering** received a **CA$30 million** contract to design and implement the *engine intake and exhaust system* and *infrared (IR) suppression devices*. The contract supports jobs in Ottawa [10] .	Irving Shipbuilding article [10] .
Flexible hose systems and bellows	**Patlon Aircraft & Industries**, an Ontario company, was awarded contracts (worth >CA$30 million) to design and install *flexible hose systems and bellows*, supporting local manufacturing and workforce development [11] .	Patlon article [11] .
Nuts, bolts and fasteners	**No publicly-announced contract** identifies a single supplier for fasteners. The RCD's supply chain uses general industrial fastener distributors in Canada and internationally. Companies such as **BW Industrial Sales, Pacific Bolt** and **Cardinal Fasteners** supply military-grade nuts, bolts, screws and washers to manufacturing programs. However, official statements emphasize that major structural components, not fasteners, are domestically sourced; thus fasteners are likely procured from multiple vendors rather than a single contract.	

Propulsion and power systems

The River-class employs a **CODLOG (Combined Diesel-Electric or Gas)** arrangement. Each ship uses diesel generators and electric motors for normal operations and a gas turbine for high-speed bursts. The supply chain is multinational but includes significant Canadian content.

System	Supplier & manufacturing location	Evidence
Primary gas turbine	**Rolls-Royce MT30 gas turbine**. Rolls-Royce will supply a 36-MW MT30 gas turbine for each ship; the contract was signed with Irving Shipbuilding. Rolls-Royce's May 2025 release confirms the MT30 supply [12] . MT30 modules are produced at Rolls-Royce facilities in the UK and U.S.; they will be delivered to Halifax for integration.	Rolls-Royce press release [12] .

System	Supplier & manufacturing location	Evidence
Diesel generators	**Rolls-Royce mtu Series 4000** (16-cylinder diesel sets). Rolls-Royce's release notes that mtu Series 4000 diesel generators will be license-built in Canada by **Wajax Power Systems**, ensuring domestic manufacturing [12].	Rolls-Royce press release [12].
Electric propulsion motors	**General Electric (GE) Power Conversion**. The RCD uses two high-power electric motors similar to those on the UK Type 26 frigates. A Navy Lookout article explains that GE's plant in **Rugby, United Kingdom** produces ultra-quiet motors used in the Type 26 and that Canada's CSC/River-class relies on this facility [13]. The article warns that the motors can only be built in Rugby and that closing the plant would affect the Canadian program [14].	Navy Lookout analysis [13] [14].
Mission Bay Handling System	See structural materials (Rolls-Royce MBHS).	Rolls-Royce press release [8].
Propellers and steering gear	**Kongsberg Maritime**; details above.	Naval Today [6].

Sensors, radar and electronic systems

System	Supplier & manufacturing location	Evidence
Combat Management System	**Lockheed Martin Canada (LMC)** will provide the **CMS 330** combat management system. LMC leads the design and integration of sensors and weapons; subcontractors include BAE Systems, CAE, MDA, L3Harris, Ultra and others, with engineering work carried out in Canada [15].	National Defence project summary [15].
Primary radar	**AN/SPY-7(V)3 (SPY-7) radar**. In December 2025, the U.S. Department of Defense awarded Lockheed Martin a **US$104.6 million** foreign military sales contract to produce SPY-7 radars for Canada. Work will be performed at Lockheed Martin's **Moorestown, New Jersey facility** with completion by January 2032【181464457829690†L59-L81】. The radar is derived from Japan's Aegis Ashore system and provides long-range, multi-mission coverage.	
Laser warning and countermeasure system	**MDA** (formerly MacDonald, Dettwiler and Associates). MDA is responsible for the **Laser Warning & Countermeasure System** and integration of the electronic warfare suite. A 2024 MDA release notes that the system will be developed and built at MDA facilities in **Richmond (BC)**, **Brampton and Kanata (Ontario)**, **Montréal (Quebec)** and **Dartmouth (Nova Scotia)** [16] [17].	

System	Supplier & manufacturing location	Evidence
Electronic warfare suite	Canada decided in 2024 to drop a bespoke EW suite and instead purchase the **AN/SLQ-32(V)6** electronic warfare system via U.S. Foreign Military Sales. The Journal of Electromagnetic Dominance reported that the decision would increase commonality with U.S. Navy ships; the system integrates **BAE Systems' Nulka** active decoy, while MDA's laser warner remains [18] . Production of the AN/SLQ-32(V)6 occurs at U.S. facilities; integration will be done by LMC in Canada.	
Hull-mounted sonar	**Ultra** will supply the **S2150-C** hull-mounted sonar, derived from the Type 26 frigate sonar. Ultra's contract includes technology transfer and is expected to create high-tech jobs in Canada [19] . Ultra has Canadian facilities (e.g., Dartmouth, NS) where significant work will occur.	
Towed array sonar	**Thales S2087 (CAPTAS 4)**. In May 2026 Lockheed Martin Canada awarded Thales Canada a contract to supply the S2087 low-frequency variable-depth towed array sonar. Thales said it will work with Canadian industry to perform maintenance and upgrades domestically [20] [21] . The sonar modules are produced in France; integration and through-life support will involve Canadian partners.	
Torpedo launchers	**SEA Canada (part of SEA Group)** will manufacture the lightweight torpedo launcher systems at its **new Ottawa facility**. SEA's 2026 release notes that the contract provides integrated logistic support and ensures the systems are *Canadian-built* [22] .	
Visual Surveillance System & Integrated Platform Management System	**L3Harris Technologies** supplies the **Integrated Platform Management System (IPMS)** and **Control & Instrumentation** and will also deliver a *Visual Surveillance System (VSS)*—a network of cameras providing 360° coverage of the ship. An Irving Shipbuilding news release states that the VSS is developed and manufactured at **L3Harris' facility in Montréal** and brings the total value of L3Harris work on the first three ships to over CA$170 million [23] .	
Integrated bridge and navigation system	**OSI Maritime Systems** provides the **Integrated Bridge and Navigation System (IBNS)**. Lockheed Martin Canada awarded OSI the implementation contract for the first three ships, and OSI stated that the systems are designed and manufactured at its **Burnaby, British Columbia headquarters** [24] .	

System	Supplier & manufacturing location	Evidence
Communications and sensor suites	Additional systems include **L3Harris communications and datalinks**, **General Dynamics sonobuoy processing**, **BAE Systems optical and infrared sensors**, **CAE training systems** and **Kongsberg Naval Strike Missile fire-control systems**. These are integrated into the CMS 330 and involve multiple Canadian facilities; official sources emphasise domestic participation but have not publicly detailed each production site [15].	

Weapons and combat systems

Weapon or system	Supplier & sourcing	Evidence
Main gun	The RCD will carry a **127 mm/5-inch Mk 45 Mod 4 naval gun** from **BAE Systems**. While BAE Systems has not publicly released a dedicated news release on the Canadian contract, the gun is built at BAE's Louisville, Kentucky facility where all Mk 45 guns are manufactured. Integration into the Canadian ships will occur at Irving Shipbuilding.	
Surface-to-air & anti-ballistic missiles	The RCD will employ **Raytheon SM-2 Block IIIC Standard Missiles** and **ESSM Block II** as part of its Aegis-derived air-defence suite. Procurement occurs via **U.S. Foreign Military Sales**; missiles are produced at Raytheon facilities in the U.S. The RCD will also be Tomahawk-capable; Canada signed a letter of offer with the U.S. Navy to acquire Tomahawk cruise missiles in 2023.	
Anti-ship missiles	**Kongsberg Naval Strike Missile (NSM)**. Canada intends to arm the River-class with NSM; production of NSM missiles occurs in Norway and the U.S., while integration on the ships is handled by LMC.	
Torpedo launchers	See sensors section: SEA Canada manufacturing [22].	
Decoy and countermeasures	**BAE Systems Nulka** active missile decoys will be integrated with the AN/SLQ-32(V)6 electronic warfare system [18]. BAE's Nulka rockets are built in Australia, while launchers will be installed on the RCD in Canada.	
Small arms and close-in weapons	The RCD will carry **Phalanx CIWS** from Raytheon and remote-controlled weapon stations; these are purchased through FMS and integrated by LMC.	

Industrial and economic impact

- The **implementation contract** for the first three ships is worth **CA$8 billion** (including taxes), covering construction and initial support; the projected total cost to build and deliver these three ships is **CA$22.2 billion** [25] . The investment is part of Canada's National Shipbuilding Strategy aimed at revitalizing domestic shipbuilding.
- The program supports **5,250 jobs annually** and contributes roughly **CA$719 million to GDP per year** [26] . Additional spending by employees supports another **1,545 jobs** and **CA$191 million** in GDP [27] . These benefits are concentrated in **Atlantic Canada**, particularly around Halifax where Irving Shipbuilding's yard is located [28] .
- Numerous **Canadian small and medium enterprises** (SMEs) are subcontractors. Companies like **Marswell Metal**, **W.R. Davis Engineering**, **Patlon**, **SEA Canada**, **OSI Maritime Systems**, **MDA** and **L3Harris Canada** provide components ranging from ballast to sophisticated sensors, ensuring a domestic supply chain and technology transfer.

Conclusion

The River-class destroyer program is the largest naval procurement in modern Canadian history. It uses a **mixed international-Canadian supply chain**: heavy structural components such as steel, ballast and mission bay handling systems are produced in Canada; high-tech propulsion (electric motors and gas turbines) is largely imported from the UK and U.S.; sensors and radar involve a combination of U.S. technology (SPY-7, AN/SLQ-32) and Canadian design (MDA laser warner, Ultra sonar); and weapons systems are mostly acquired through U.S. Foreign Military Sales. The program emphasises **technology transfer and domestic manufacturing**, with many subsystems being integrated or built at Canadian facilities in Halifax, Ottawa, Montréal, Burnaby and other cities. This approach ensures that Canada not only receives modern warships but also develops a sovereign shipbuilding capability and industrial base for future naval programs.

---

[1] [2] [25] [26] [27] [28] Government of Canada announces contract award for the construction of the River-class destroyers for the Royal Canadian Navy - Canada.ca
https://www.canada.ca/en/department-national-defence/news/2025/03/government-of-canada-announces-contract-award-for-the-construction-of-the-river-class-destroyers-for-the-royal-canadian-navy.html

[3] [4] Irving Shipbuilding Awards Metals Supply Contract to Russel Metals, Supporting Jobs and Communities in Nova Scotia - Irving Shipbuilding
https://shipsforcanada.ca/our-stories/irving-shipbuilding-awards-metals-supply-contract-to-russel-metals-supporting-jobs-and-communities-in-nova-scotia

[5] Marswell Metal Supports River-Class Destroyer Shipbuilding
https://marskeel.com/marskeel-marswell-irving-shipbuilding-destroyer-program/

[6] Kongsberg to supply propulsion systems for Canada's River-class destroyers - Naval Today
https://www.navaltoday.com/2025/09/09/kongsberg-to-supply-propulsion-systems-for-canadas-river-class-destroyers/

[7] Kongsberg Maritime to supply fixed-built propellers for River-class destroyers
https://www.janes.com/defence-intelligence-insights/defence-news/defence/kongsberg-maritime-to-supply-fixed-built-propellers-for-river-class-destroyers

[8] Rolls-Royce awarded Mission Bay Handling System contract for Type 26 frigates | Rolls-Royce
https://www.rolls-royce.com/media/press-releases/2024/13-08-2024-rr-awarded-mission-bay-handling-system-contract-for-type-26-frigates.aspx

[9] Curtiss Wright Awarded 25M Contract To Provide Aircraft Handling Systems AND Support FOR Royal Canadian Navy River Class Destroyers | EMS | Curtiss-Wright
https://electro-mechanicalsystems.curtisswright.com/news/news-details/2025/Curtiss-Wright-Awarded-25M-Contract-To-Provide-Aircraft-Handling-Systems-And-Support-For-Royal-Canadian-Navy-River-Class-Destroyers

[10] Ottawa's W.R. Davis Engineering awarded $30-million supply chain contract to support the River-class destroyer project (Canadian Surface Combatant) for Canada's Navy - Irving Shipbuilding
https://shipsforcanada.ca/our-stories/ottawas-w-r-davis-engineering-awarded-30-million-supply-chain-contract-to-support-the-river-class-destroyer-project-canadian-surface-combatant-for-canadas-navy

[11] Patlon Secures $30M Contract with Irving Shipbuilding for River-Class Destroyer Project. - Patlon
https://patlon.com/patlon-secures-30m-contract-with-irving-shipbuilding-for-river-class-destroyer-project/

[12] Rolls-Royce to provide power, handling system for Canadian River-class destroyers | Rolls-Royce
https://www.rolls-royce.com/media/press-releases/2025/28-05-2025-rr-to-provide-power-handling-system-for-canadian-river-class-destroyers.aspx

[13] [14] Closure of GE Rugby electric motor plant threatens supply of Royal Navy propulsion systems - Navy Lookout
https://www.navylookout.com/closure-of-ge-rugby-electric-motor-plant-threatens-supply-of-royal-navy-propulsion-systems/

[15] River-class Destroyer Project - Canada.ca
https://www.canada.ca/en/department-national-defence/services/procurement/canadian-surface-combatant.html

[16] [17] MDA Awarded First Production Contract for the Canadian Surfa | MDA Space
https://mda.space/article/mda-awarded-first-production-contract-canadian-surface-combatant

[18] Canada Looks to US Navy for Non-Developmental EW for CSC – Journal of Electromagnetic Dominance
https://www.jedonline.com/2024/10/22/canada-looks-to-us-navy-for-non-developmental-ew-for-csc/

[19] Ultra to provide hull-mounted sonar for Canada's CSC project
https://www.naval-technology.com/news/ultra-hull-mounted-sonar-canadas-csc-project/

[20] Thales, Lockheed Martin to provide towed array sonar for Canada's River-class destroyers - Naval Today
https://www.navaltoday.com/2026/05/29/thales-lockheed-martin-to-provide-towed-array-sonar-for-canadas-river-class-destroyers/

[21] Canada Selects Thales Anti-Stealth Sub Sonar for River-Class Destroyers
https://thedefensepost.com/2026/06/01/canada-thales-sonar/

[22] Lockheed Martin Canada Awards SEA Canada with Implementation Phase Contract for River-class Destroyers
https://www.sea.co.uk/news/lhm-piphase

[23] Irving Shipbuilding awards L3Harris new River-class destroyer contract: Visual Surveillance System - Irving Shipbuilding
https://shipsforcanada.ca/our-stories/irving-shipbuilding-awards-l3harris-new-river-class-destroyer-contract-visual-surveillance-system-2

[24] Latest Updates From OSI
https://osimaritime.com/lmc-river-class-destroyers/